TrueDeed: Algorithmic Land Fraud & Double-Selling Prevention Platform

Official Concept Note & Technical Proposal — DIU LandTech Innovation Hackathon 2026

Team Code: LA-72 (Team TrueDeed)	**Track:** Digital Land Records, Verification & Digitization	**Selected Problem:** P3: Catch Fraud & Double-Selling

1. Problem Chosen: P3 — Catch Fraud & Double-Selling

Under Section 5 of the LandTech 2026 Rule Book, Team TrueDeed selects **Problem P3: Catch Fraud & Double-Selling**. Land disputes are the single largest burden on Bangladesh's judicial and civil administration, and while portals like e-Porcha and e-Namjari digitize historical records, they are passive repositories with no real-time verification gate at the Sub-Registrar desk. Because deed registration (Ministry of Law) and mutation (Ministry of Land) are disconnected, fraudsters exploit the gap to over-sell inherited shares, fabricate intermediate Baya deeds, and double-sell land before records are updated.

2. Why It Matters in Bangladesh (National Urgency & Evidence)

Land disputes drive social friction, generational litigation, and court congestion in Bangladesh. Per Law Commission and Supreme Court data, property and boundary disputes account for roughly 70% of the ~3.5 million pending civil cases, locking up billions of takas in productive wealth.

The Core Institutional Void: Deed registration (Directorate of Registration, Ministry of Law) and Record-of-Rights/Mutation (AC Land offices, Ministry of Land) run separately, with no real-time synchronization at deed execution. Fraud syndicates exploit this in three ways:

1. **Inheritance Over-Selling:** Heirs conceal female siblings or forge heir certificates and sell 30-40 decimals from an estate where their lawful Farayez share is only 15.
2. **Broken Baya Lineage:** Fabricated intermediate deeds with fictitious sub-registry volume numbers break the CS → SA → RS chain of title, and buyers cannot verify them manually.
3. **45-Day Double-Selling Window:** Mutation takes 30-45 days after registration, during which records still show the original seller, who can sell the same plot to multiple buyers.

The Breadth of Harm:

1. **Judicial Burden:** Land cases congest sub-district and district courts for decades.
2. **Banking Risk:** Banks and NBFIs spend 14-21 days manually vetting titles, yet still suffer NPLs from forged Baya deeds.
3. **Vulnerable Heirs:** Widows, daughters, and minors are deprived when deeds exceed the legal Farayez quota.
4. **Diaspora Security:** Over 10 million Non-Resident Bangladeshis face ancestral land-grabbing risk with no remote, authenticated title-monitoring portal.

3. Proposed Solution: The TrueDeed Infrastructure

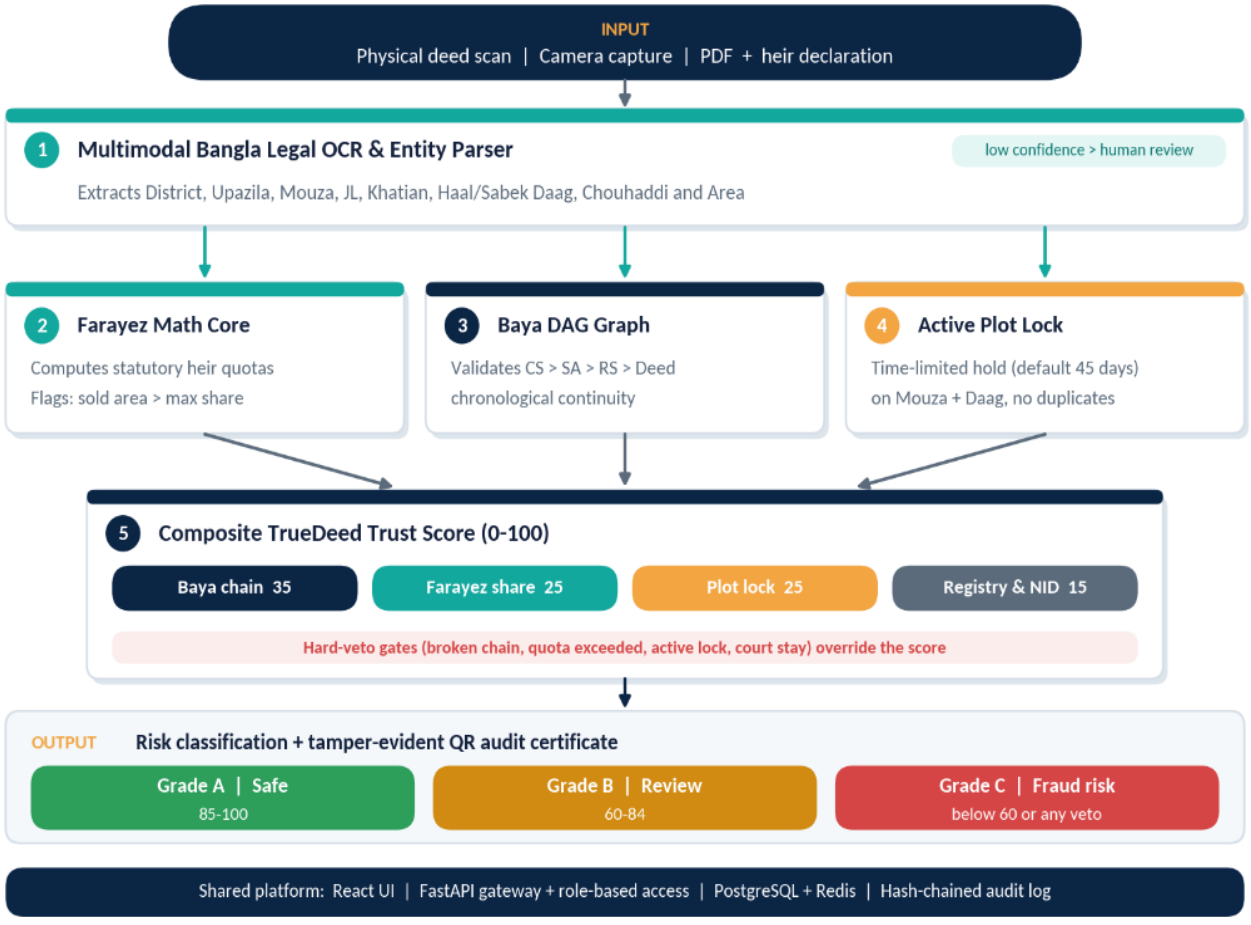
TrueDeed is an Algorithmic Decision Support System (ADSS) that bridges the gap between Sub-Registry and AC (Land) offices. Unlike passive portals such as e-Porcha, it acts as an active pre-

execution fraud gate, checking title integrity, inheritance quotas, and transaction locks in under 5 seconds, before a deed is executed. It is deployed at Sub-Registry desks, bank credit-vetting portals, and citizen self-service channels, and rests on three pillars:

1. **Inheritance Math Engine:** Calculates each heir's lawful Farayez share and flags any sale that exceeds it (e.g., 30-40 decimals against a lawful 15).
2. **Baya Lineage Graph Tracer:** Uses a Directed Acyclic Graph (DAG) to trace ownership back to CS/SA/RS records, so fabricated intermediate deeds fail verification.
3. **Plot Concurrency Lock:** Places a temporary lock on the plot during registration, blocking duplicate filings across offices until mutation is complete.

4. How It Works: System Architecture & WorkFlow

TrueDeed transforms physical deed scans into mathematical proofs of title authenticity via a modern 5-stage microservice architecture:



5. Core Technology Stack & Implementation Feasibility

1. **Frontend Portal (React 19, Tailwind CSS, TypeScript):** A fast, responsive web dashboard for Sub-Registrar desks, bank vetting officers, and citizen self-service.
2. **API Gateway (FastAPI / Python 3.12, Uvicorn, Celery):** An asynchronous microservice orchestrator with automated request routing, auth tokens, and audit logging.

3. **Bangla Legal OCR (Gemini Flash VLM, OpenCV Pipeline):** An asynchronous pipeline (3-5s) that extracts District, Mouza, Khatian, Daag, Chouhaddi, and acreage into JSON.
4. **Farayez & DAG Engine (NetworkX, Custom Math Core):** Sub-second (<500ms) graph traversal of historical Baya deeds and mathematical calculation of statutory quotas.
5. **Database & Ledger (PostgreSQL 16, Redis, SHA-256 Hashes):** ACID-compliant storage of Khatians, 45-day active plot locks, and an append-only cryptographic audit trail.

6. Core Functional Modules

TrueDeed delivers five targeted innovations to resolve Problem P3 at the national level:

1. **Farayez Quota Interceptor:** Computes each heir's lawful share (including *Aul* and *Radd*) and blocks any sale that exceeds the seller's entitlement.
2. **DAG Baya Lineage Tracer:** Rebuilds the chain of title from CS/SA/RS surveys to the current seller, flagging fake deeds, missing mutations, and broken continuity.
3. **45-Day Plot Concurrency Lock:** Holds the target Daag record across all registration desks until e-mutation completes, preventing double sales.
4. **Bangla Legal OCR and Normalizer:** Converts handwritten and archaic Bengali deed scans into structured JSON (Mouza, Khatian, Daag, Chouhaddi, acreage), with human-in-the-loop verification.
5. **Composite Trust Score (0-100%):** An explainable, rule-based security score with a tamper-evident, QR-verifiable digital certificate.

7. TrueDeed Composite Trust Score Matrix (0–100%) & Hard Veto Rules

Each transaction is scored out of 100 across four checks:

1. **Baya Chain Continuity (35):** Unbroken DAG lineage from CS/SA/RS to the current seller.
2. **Farayez Quota Compliance (25):** Transferred acreage within the seller's statutory entitlement.
3. **Active Plot Lock Status (25):** No concurrent filings or active locks on the Daag within 45 days.
4. **NID & Registry Consistency (15):** Seller NID, Porcha records, and biometric logs cross-verified.

Hard vetoes: A broken Baya chain drops the score below 30% (fatal block). An active civil court injunction (Order 39, Rules 1-2 CPC) drops it to 0% and locks the transaction.

8. Nationwide Deployment & E-Governance Adoption Strategy

- **Sub-Registry Desks (Ministry of Law):** Integrated directly into the deed registration software at Sub-Registry desks to verify ownership and statutory heir quotas before printing stamp papers.
- **AC Land Offices (Ministry of Land):** Automates cross-checking during e-Mutation hearings, instantly detecting forged Baya deed submissions and duplicate applications.
- **Commercial Banks & Mortgage Lenders:** Integrated via secure B2B REST API into core banking loan origination software, collapsing collateral property title vetting from 21 days down to 5 seconds.
- **Citizens & Expatriate Investors:** Mobile-responsive citizen portal ('Check Before You Pay') enabling prospective buyers and expatriates to verify title validity before paying token money.

9. Team Composition & Mentors (Rule Book Compliance)

In strict compliance with Section 6 of the LandTech 2026 Rule Book, Team TrueDeed comprises 2 DIU students and 2 specialized mentors:

Role	Name	Affiliation/ID	Contact/Responsibility
Student 1 (Team Leader)	Ishtiak Ahmed	ID: 0242310005101006 Dept. of Computer Science and Engineering, DIU	System Architecture, Full-Stack Engine & Technical Defense
Student 2 (Core Member)	Ashfikur Rahman Mahi	ID: 0242310005101618 Dept. of Computer Science and Engineering, DIU	UI/UX Architecture, Concept Documentation & Quality Assurance
Faculty Mentor	Md. Jubayar Alam Rafi	Lecturer, Department of CSE Daffodil International University	alamrafi.cse@diu.edu.bd Phone: 01707587089
Industry Support Mentor	MD Zafor Iqbal	CSE-TECH	xoy4444@gmail.com GovTech Feasibility & Enterprise Scaling

10. Hackathon Execution Roadmap & Safeguards

- **Phase 1 (Current Submission):** Concept Note submission, Formal Technical Architecture Blueprints, and Live GitHub Repository showcase.
- **Phase 2 (Final On-Site Hackathon):** Live interactive prototype demonstration featuring real-time deed upload, asynchronous OCR parsing, Farayez fraud interception, and DAG timeline rendering.
- **National Deployment Horizon:** Collaborative sandbox pilot with the Ministry of Land and Sub-Registry offices under national e-governance innovation channels.